MINOR PROJECT

Development of Digital Mental Health and psychological support system for students in Higher Education

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INTRODUCTION

Mental health affects how students think, feel, and act and how they handle stress and make decisions.

Why It Matters

- Mental health impacts focus, grades, and overall well-being.
- Early digital support can prevent severe issues and dropouts.
- Technology makes help easy, timely, and stigma-free.

Stats

- **1 in 7 students** face mental health issues (WHO)
- **60% of college students** report stress or anxiety
- Only **1 in 4** seek help

PROBLEM STATEMENT

Many students in higher education experience stress, anxiety, and depression but lack timely access to mental health support. Due to stigma, limited counselors, and academic pressure, most do not seek help.

Traditional counseling is often limited by stigma, cost, and accessibility (Hunt & Eisenberg, 2010).

There is a need for a **digital platform** that provides **confidential, accessible, and continuous psychological support** to help students manage their mental well-being effectively.

However, most existing systems **aren't tailored for higher education students**.

Our digital platform can bridge this gap through **personalized and campus-focused support**

OBJECTIVES

- To design a smart, user-friendly **digital platform** to support the emotional and mental well-being of students through AI guidance and self-help tools.

- To offer **24/7 access** to self-assessment tools and emotional assistance.
- To ensure **confidential and stigma-free** communication for students.
- To **connect students with counselors** or professional help when needed.
- To promote **early detection and prevention** of mental health issues
- To help students track and manage stress/anxiety levels

PROPOSED SOLUTION

✅ **How the System Works**

- **User Logs In**
  Students access the platform anonymously.
- **AI Mental Health Assistant Interacts**
  AI chatbot provides initial support, listens to concerns, and identifies stress levels.
- **Personalized Recommendations**
  Based on user input, the system suggests mood tracking, self-help exercises, or coping resources.
- **Access to Services**
  Students can choose:
  - **Anonymous Counselling**
  - **Self-Help Resources**
  - **Well-Being Dashboard**
  - **Community Support**
- **Crisis Detection**
  If severe distress is detected, the system triggers alerts/helplines.
- **Progress Tracking**
  The platform monitors improvement through mood logs, habits, and daily activities.

FEATURES

- **Confidentiality & Privacy** → Anonymous login + end-to-end encryption ensures stigma-free usage.
- **Early Detection of Mental Health Issues** → AI-powered self-assessment quizzes with instant feedback.
- **Continuous Emotional Support** → AI Chatbot (24/7) provides immediate guidance in multiple Indian languages.
- **Crisis Management** → “Help Me Now” button links students to suicide prevention helplines, emergency contacts, and the option to book virtual therapy sessions instantly.
- **Habit Building & Motivation** → Gamified streaks, rewards, and badges encourage students to build healthy routines (journaling, meditation, no-substance days).
- **Holistic Student Wellness** → Covers a wide spectrum of problems: academic pressure, career uncertainty, financial stress, addiction, relationship issues, family expectations, social media pressure, past trauma, and identity/sexuality support.
- **Micro-Content Library** → Reels-style content for quick reading such as on financial tips (2-3 minutes) instead of long articles

LITERATURE REVIEW

Comparison with Existing Systems :

Feature	Our App	Headspace	Sanvello	Talkspace
🎯 Student-focused	✅	❌	❌	❌
📝 Self-assessments	✅	❌	✅	❌
📊 Mood & progress tracking	✅	✅	✅	❌
👥 Peer support	✅	❌	❌	❌
🩺 Licensed counsellor sessions	✅	❌	❌	✅
🚨 Crisis intervention (hotlines)	✅	❌	❌	❌
📈 Admin dashboard (for campuses)	✅	❌	❌	❌

METHODOLOGY

Tech Stack:

Frontend

Flutter React

Backend

express

faśIPAI WebRT

Backend

PostgreSQL

mongoDB

AI / NLP

Hugging Face

OpenAI

Telehealth

WebRTC

Twilio

Notifications

Firebase

SendGrid

Infrastructure

Docker

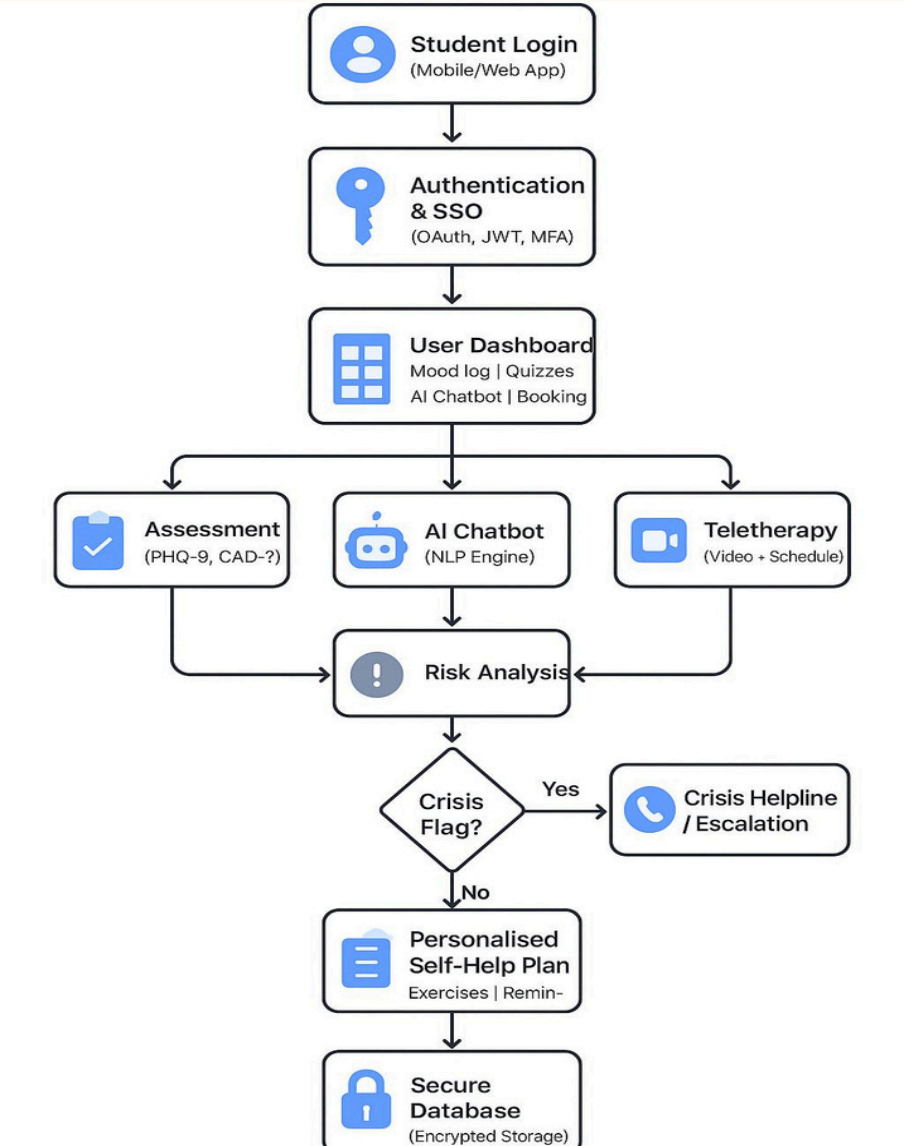
Kubernetes

Security

JWT TLS

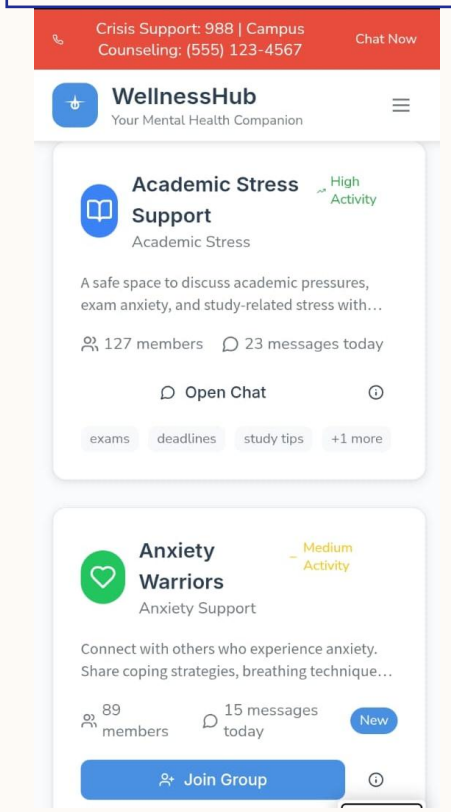
OWASP

User Flow Diagram:

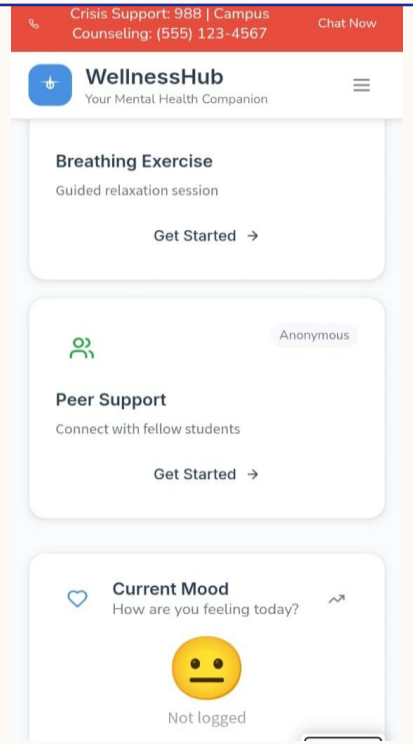


IMPLEMENTATION

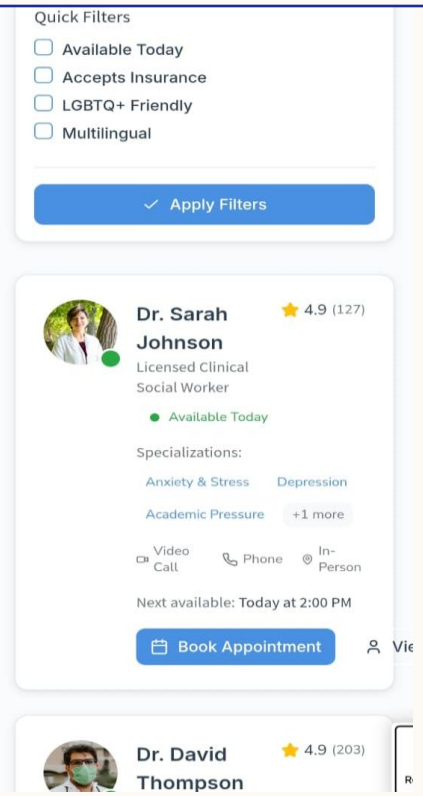
Peer Support



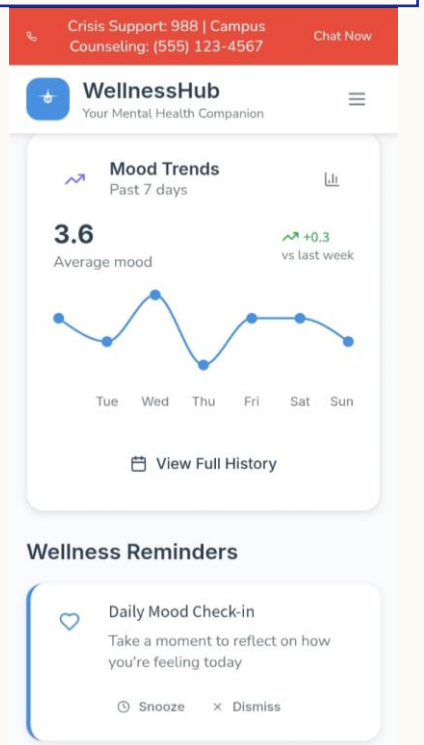
Coping Resources



Counsellor Support

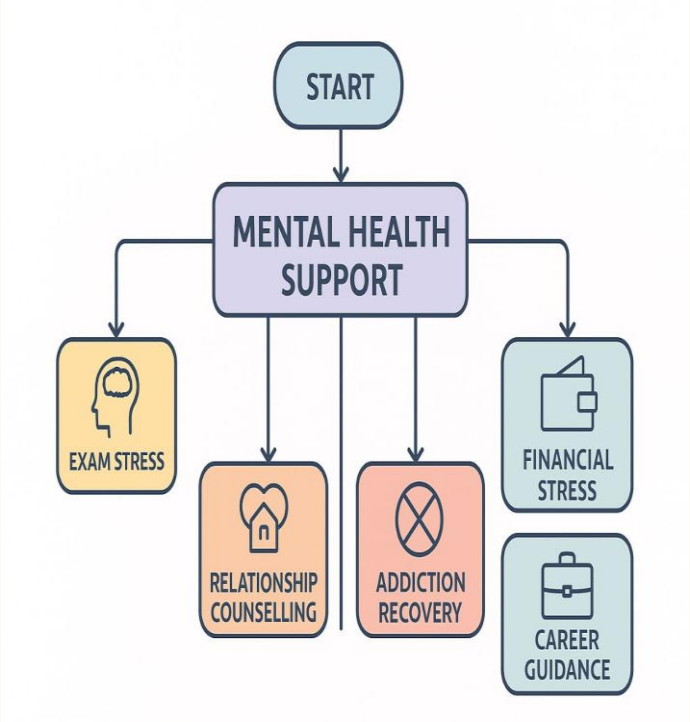


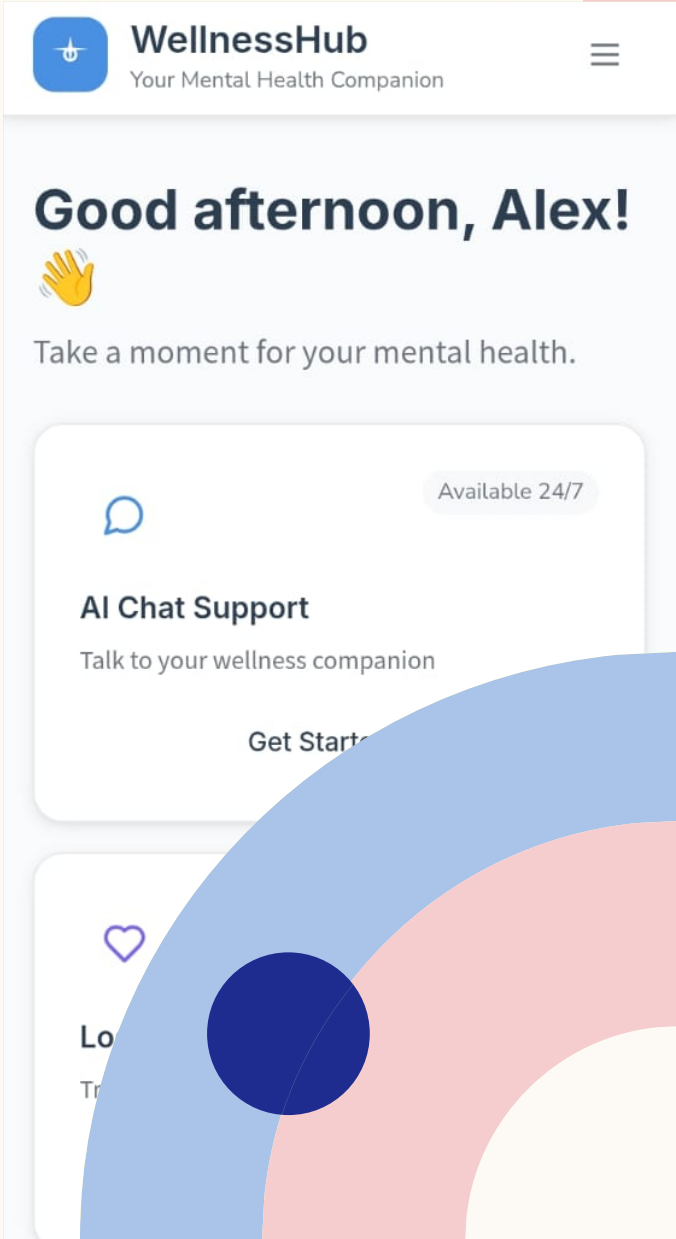
Mood Tracking



IMPACT & BENEFITS

- ✅ Improved Student Well-Being
- ✅ Personalized Support for Every Student
- ✅ Strengthened Campus Mental Health System
- ✅ Better Academic Performance
- ✅ Early Detection of Mental Health Issues
- ✅ Provides support for sensitive issues (identity, relationships, trauma).
- ✅ Increases student retention rates, saving financial losses due to dropouts.





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Q&A

THANK YOU!